

BENCHMARKING INSTANCE SEGMENTATION ARCHITECTURES: FAILURE MODES AND IMPROVEMENTS

•Team 16

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INTRODUCTION

- Instance segmentation predicts a separate pixel mask for each object
- It combines detection, classification, and boundary prediction
- It is harder than classification because object shape must be precise

Image source: <https://3dvar.com/He2017Mask.pdf>



BACKGROUND

Mask R-CNN represents
proposal-based CNN
segmentation

DETR represents
transformer-based object-
query segmentation

SAM represents
promptable vision-
foundation segmentation

PROBLEM STATEMENT

- Small objects are often missed or poorly segmented
- Thin objects lose fine boundaries and internal gaps
- SAM performs very differently with and without good prompts

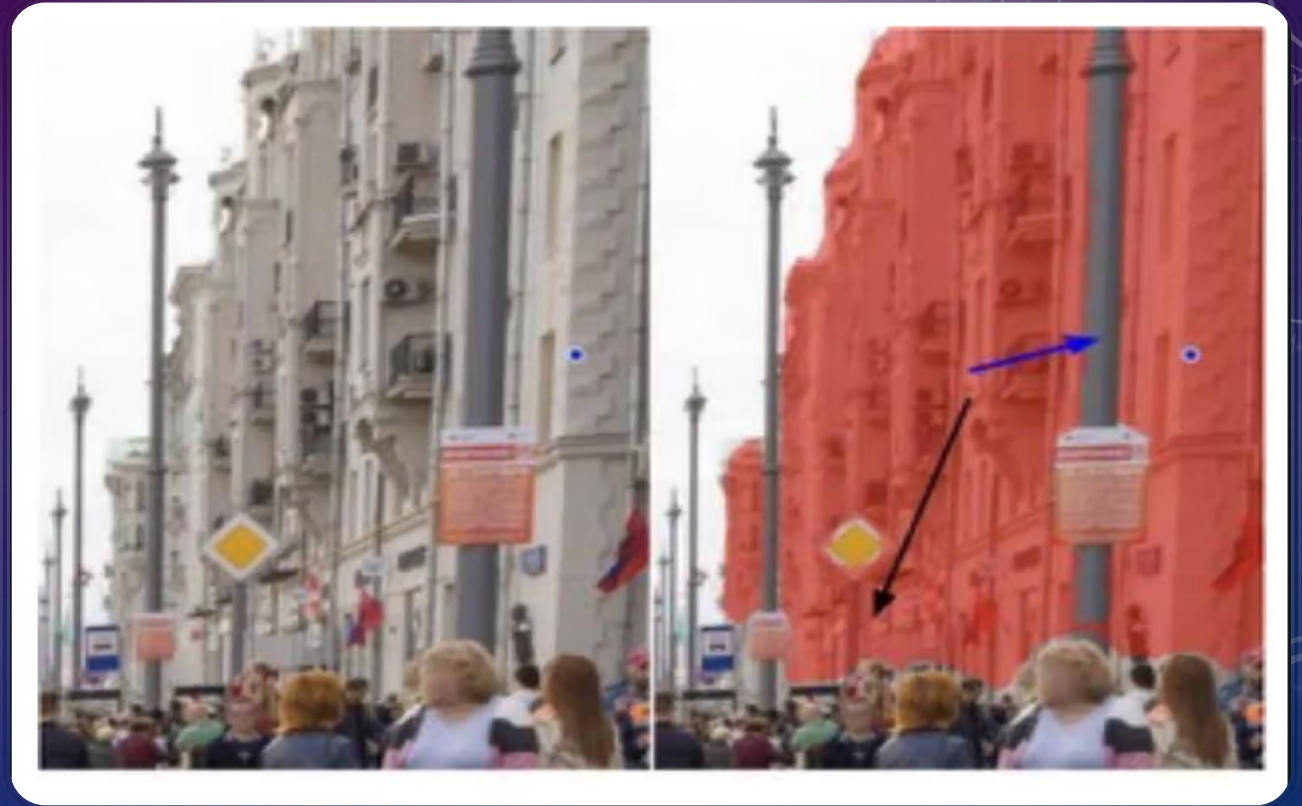


Image source: <https://www.louisbouchard.ai/meta-sam>

MOTIVATION

We need a fair comparison
under the same COCO
setup

Analyse size wise and
category wise failure

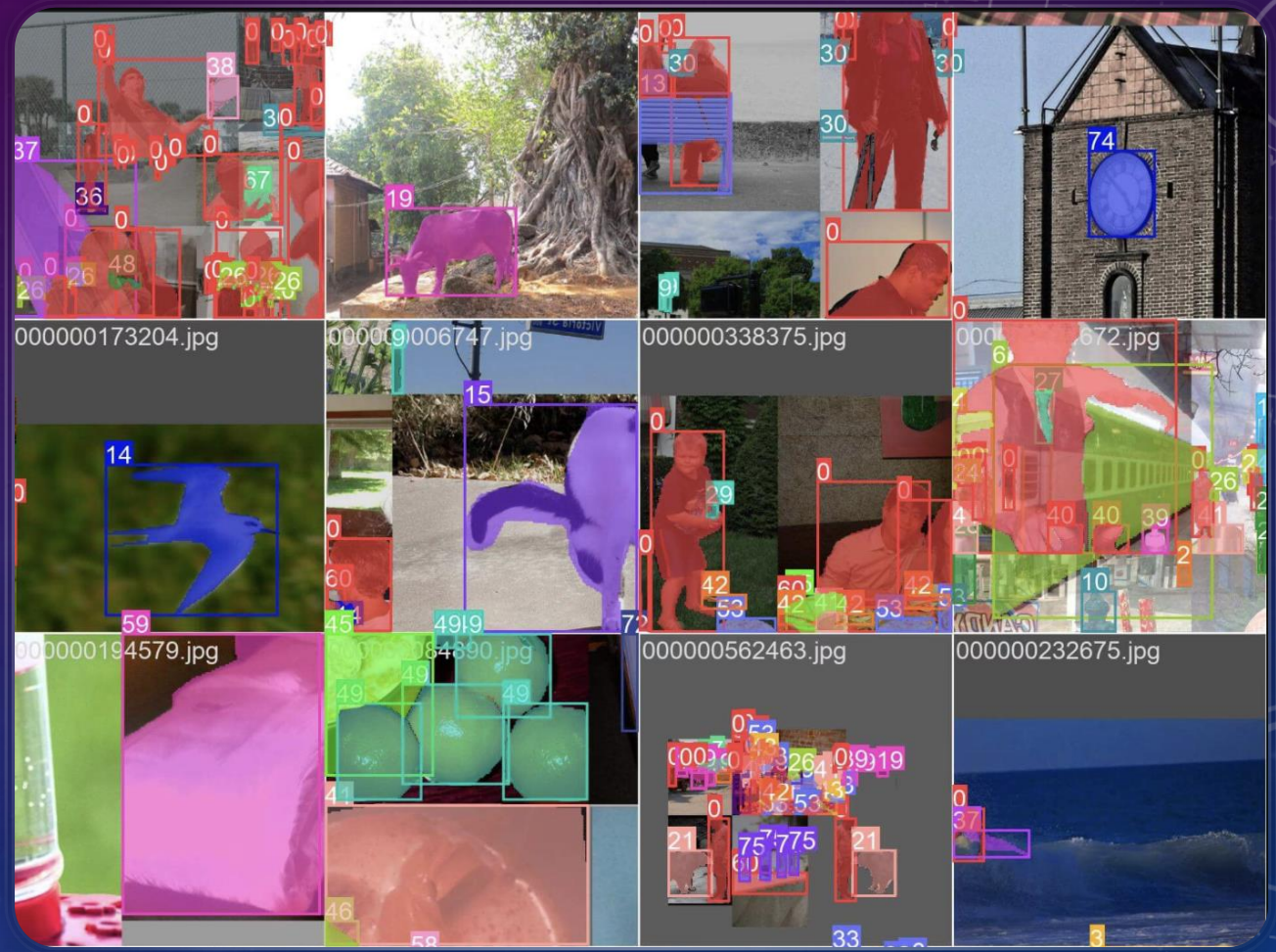
Measure the gap between
oracle SAM and realistic
SAM

DATASET AND SETUP

- Dataset: COCO val2017 balanced 200-image subset
- Total evaluated instances: 2,754 ground-truth objects
- Objects are grouped into small, medium, and large sizes
- Thin categories include fork, spoon, bicycle, chair, tie, and umbrella

Image source :

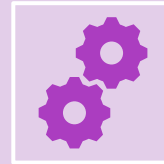
<https://docs.ultralytics.com/datasets/segment/coco#sample-images-and-annotations>



MODELS EVALUATED



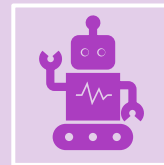
Mask R-CNN is the CNN proposal-based baseline



DETR is the transformer-based detection-segmentation model



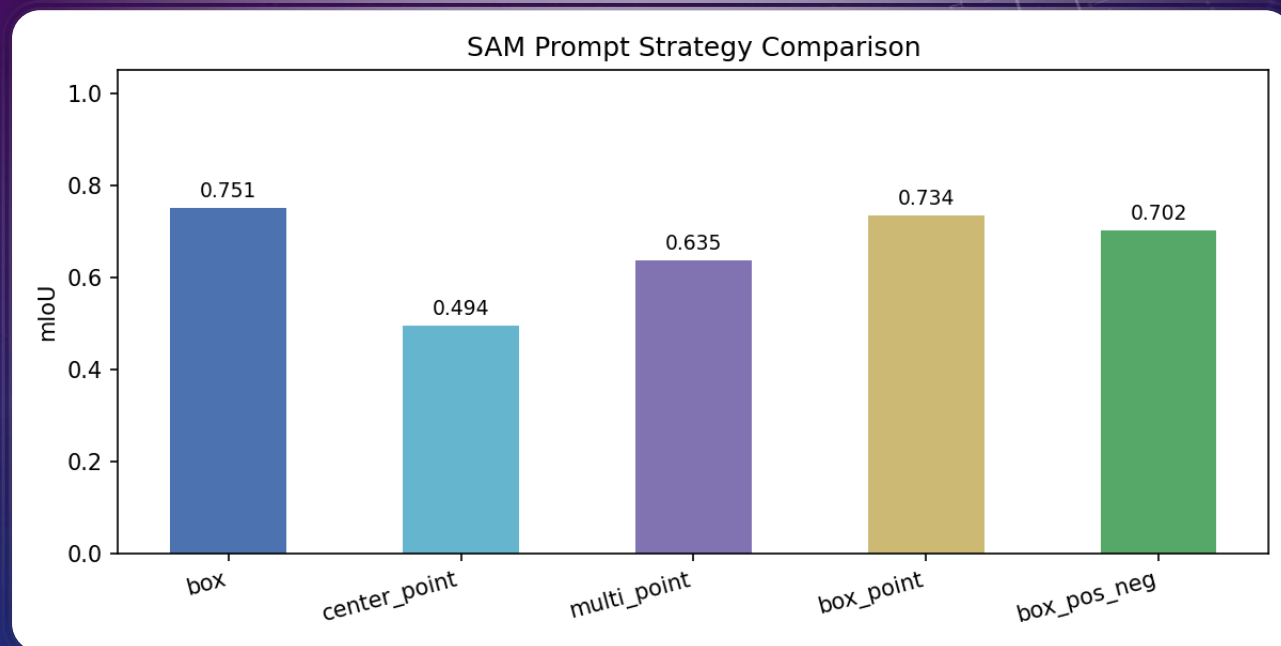
SAM is tested with five ground-truth prompt strategies



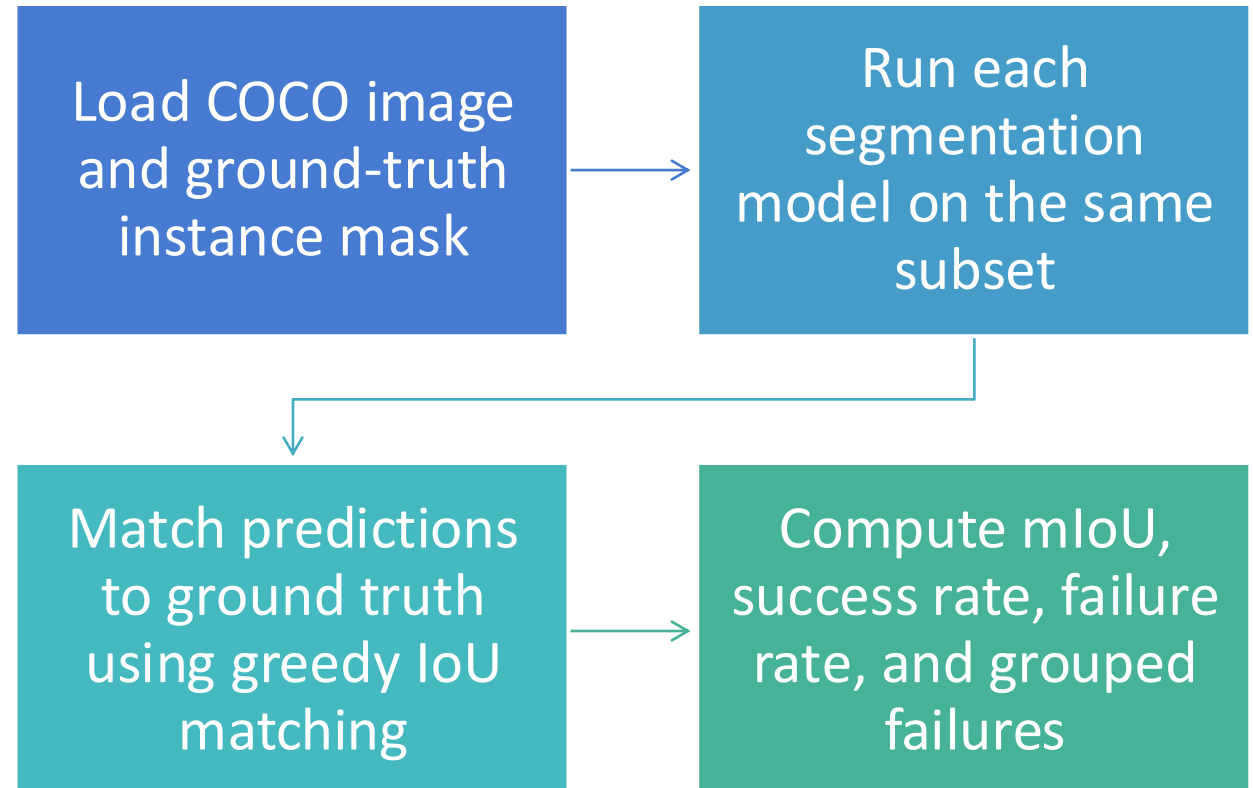
SAM-Auto and Cascade are used as improvement/deployment settings

SAM PROMPT STRATEGIES

- Box prompt uses the ground-truth bounding box
- Center-point prompt uses one point at the object centroid
- Multi-prompt uses 3-5 positive points inside the object



EVALUATION PIPELINE



EVALUATION METRICS

- IoU measures overlap between predicted mask and ground truth
- mIoU is the mean IoU over all ground-truth instances
- Success means IOU ≥ 0.5 and failure means IOU < 0.3
- It is harder than classification because object shape must be precise

Image source :

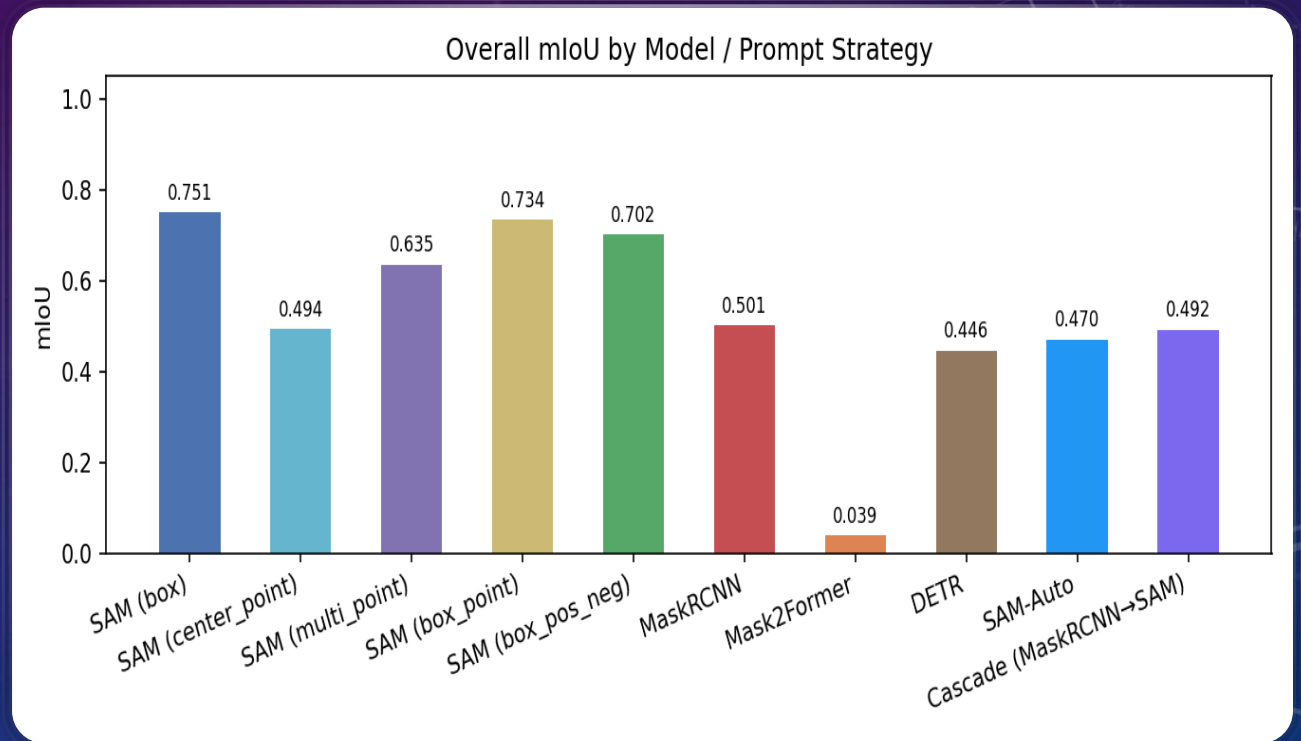
<https://pyimagesearch.com/2016/11/07/intersection-over-union-iou-for-object-detection/>

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



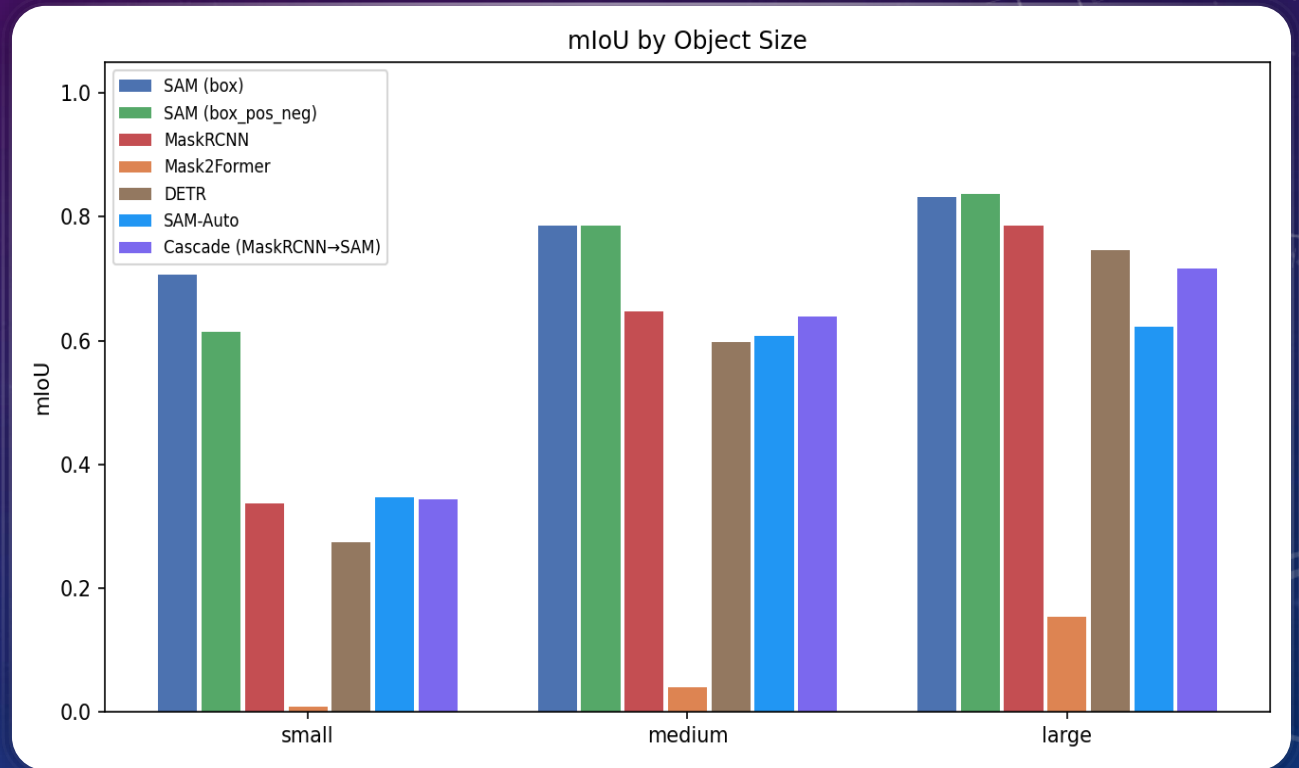
OVERALL RESULTS

- SAM with GT box prompt gives the best mIoU: 0.751
- Mask R-CNN reaches 0.501 and DETR reaches 0.446
- SAM-Auto drops to 0.470 without ground-truth prompts
- Mask2Former is excluded due to checkpoint/library calibration issue



SIZE-BASED RESULTS

- All models struggle more on small objects
- Mask R-CNN drops from 0.785 on large objects to 0.337 on small objects
- DETR drops from 0.747 on large objects to 0.275 on small objects
- SAM-box is the most stable across small, medium, and large objects



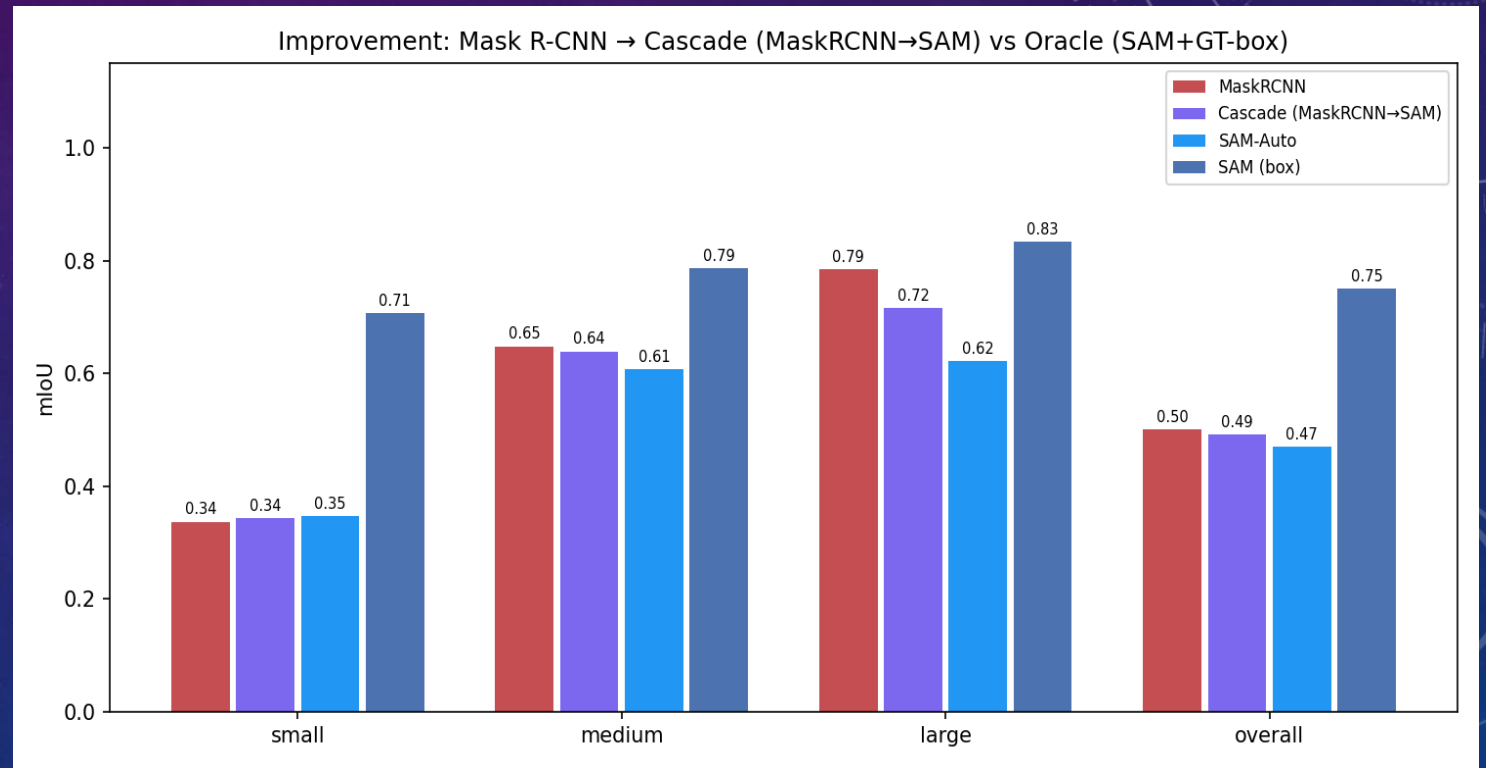
THIN OBJECT ANALYSIS

- Thin objects are harder because they require fine boundary precision
- Mask R-CNN loses detail due to its low-resolution mask head
- DETR often struggles with elongated and complex object shapes
- SAM-box reduces the thin-object gap but does not remove it fully



IMPROVEMENTS

- SAM-Auto tests realistic SAM without ground-truth prompts
- Cascade uses Mask R-CNN boxes and SAM mask refinement
- Cascade slightly improves small and thin objects
- Cascade hurts large objects, so selective use is better



CONCLUSION AND FUTURE WORK

SAM with GT box is strongest, but it is an oracle setting

Realistic SAM-Auto performs close to Mask R-CNN and DETR

Small, thin, and food objects are the main failure cases

Future work: selective cascade, better auto-prompts, and larger evaluation